

FusionCore v1 — Physics-Informed Hybrid Architecture

Stakeholder Technical Report — Stage A:

FusionCore Programme — Version 1 — Phases Ph-v1.0 through Ph-v1.6

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Executive Summary

FusionCore v1 introduces a two-branch physics-informed neural architecture, designated PiNet, with the aim of surpassing the v0 XGBoost Remaining Useful Life estimator on both regression accuracy and operational safety. PiNet integrates a temporal branch that learns degradation drift over time with a physics branch that captures the instantaneous thermodynamic state of the engine, fusing both into a shared representation that drives a life-remaining regressor and a three-class risk classifier.

Stage A training of PiNet on the pooled C-MAPSS turbofan corpus has been completed and evaluated against the NASA Official Test Set of 707 engines. The architecture failed the Critical-Recall Safety Gate by a margin of 0.5159 units of recall, recording a pooled Critical-band recall of 0.4204 against the v0 reference of 0.9363. In operational terms, Stage A PiNet misclassifies more than half of all engines that are genuinely approaching failure, a result that is not acceptable for a safety-critical maintenance decision support application.

Root-cause analysis identifies four compounding training-configuration deficiencies: an excessive learning rate that produced three-epoch convergence followed by systematic over-fitting; insufficient regularisation on the temporal branch; a training sampler that created a distribution mismatch between training and evaluation; and near-complete dominance of the regression loss term over the classification objective, leaving the risk-band head unable to learn effectively. All four causes are parametric and correctable. A Stage B retraining specification addressing each is proposed.

The v0 XGBoost baseline remains the reference operational model. No output from Stage A should be promoted to operational use. The pre-declared metric-learning ablation produced a statistically significant improvement in embedding geometry but does not resolve the headline safety failure, and its adoption is recommended only in conjunction with the Stage B parametric adjustments.

Table 1 — Headline Outcome at a Glance

Metric	v0 XGBoost (reference)	PiNet Stage A	Change
Root Mean Square Error (cycles)	14.85	24.66	+9.81 (worse)
NASA Asymmetric Score	4,336	32,204	+27,868 (7.4× worse)
Critical-Band F_2 Score	0.9339	0.4735	−0.4604
Critical-Band Recall	0.9363	0.4204	−0.5159
Safety Gate Result	Reference	FAIL	—

What these measures mean. RMSE (Root Mean Square Error) is the average error in predicted remaining life, in flight cycles; one cycle is approximately one flight. The NASA Asymmetric Score penalises over-optimistic predictions — claiming more life remains than actually does — more heavily than over-cautious ones: a 30-cycle late prediction costs roughly twice as much as a 30-cycle early one. Critical-Band Recall is the fraction of engines genuinely approaching failure that the model correctly identifies as urgent; a recall of 0.9363 means 93.6 in every 100 truly critical engines are flagged in time. The Safety Gate is the programme's binding acceptance test: it requires any candidate model to retain at least 98 per cent of the baseline's Critical-band recall before deployment can be considered.

1. Programme Context and Inheritance

1.1 What FusionCore v1 Is Attempting to Achieve

The FusionCore programme is a versioned predictive maintenance system designed to estimate the Remaining Useful Life of turbofan engines using the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) corpus [1], which models run-to-failure degradation trajectories across 709 training engines spanning four fault-mode and operational-complexity combinations. Version 0 established a competitive XGBoost baseline with a pooled Root Mean Square Error of 14.85 cycles and a Critical-band recall of 0.9363, benchmarked against the NASA Official Test Set under strict data-separation discipline. Version 1 is the first attempt to exploit a physics-informed neural architecture on the same corpus, with the hypothesis that combining thermodynamic inductive bias with learned temporal degradation trajectories will produce superior regression accuracy and improved safety-critical recall.

1.2 What Version 1 Inherits from Version 0

To ensure like-for-like comparability with the v0 baseline, the v1 programme inherits the complete v0 statistical preprocessing pipeline on a read-only basis. This inheritance covers the engine-boundary data partition that separates training, internal validation and NASA official test engines; the per-regime normalisation parameters fitted exclusively on training data; the operational regime assignments produced by the v0 regime-identification step; and the monotone transformation applied to the two heavy-tailed engineered features to make them suitable for neural network consumption¹. Two additional physics tokens, described in §2.1, are computed on top of the inherited feature set. No other preprocessing changes are introduced.

The partition sizes carried forward from v0 are shown in Table 2. The aggregate corpus yields 709 internal engines for training and validation and 707 independent test engines whose labels are withheld until the final Iron Wall evaluation.

Table 2 — Partition Sizes Inherited from v0

Partition	Engines	Windowed Observations	Role
Internal Training	567	129,331	Model weight updates
Internal Validation	142	31,028	Early stopping and checkpoint selection
NASA Official Test	707	104,897	Iron Wall headline evaluation (opened once)

¹Inheritance is read-only. No v0 statistical parameter is re-estimated in v1. This is the prerequisite for comparing PiNet against XGBoost on identical data.

2. Architecture Design

2.1 Physics-Informed Feature Augmentation

Before neural-network training, the inherited feature set is extended with two new physics-meaningful quantities that the v0 XGBoost pipeline did not compute. The first is the compressor isentropic efficiency, denoted η_c , derived from first-principles gas-turbine thermodynamics as the ratio of the ideal isentropic enthalpy rise across the high-pressure compressor to the actual measured rise [8]. This quantity directly indexes the mechanical health of the compression stage and complements the raw sensor signals by providing a thermodynamically interpretable health proxy². The second is a proportional-hazards score derived from the Cox regression model [3], which quantifies each engine-cycle observation as a risk multiplier relative to the baseline failure hazard. A Cox model is fit on the internal training cohort alone and the resulting risk score is computed for every row; it enters the architecture as a summary statistic encoding instantaneous failure proximity³.

The isentropic efficiency η_c is derived from first principles using the following expression (which appears again as Equation 1 in Appendix G, where its full derivation is given):

$$\eta_c = \frac{T_2 \left[\left(\frac{P_{30}}{P_2} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]}{T_{30} - T_2} \quad [\text{Eq. 1}]$$

This formula asks how efficiently the high-pressure compressor converts shaft power into pressure. A healthy compressor approaches the theoretical ideal; a degraded compressor requires more energy for the same pressure rise. This single derived number provides an instantaneous mechanical health index for the most failure-critical engine component.

The proportional-hazards risk score enters the feature set as Equation 2 (see Appendix G for the full specification):

$$\lambda(t | x) = \lambda_0(t) \exp(\beta^T x) \quad [\text{Eq. 2}]$$

The formula takes the current sensor readings and converts them into a single failure-risk multiplier. An engine with elevated exhaust temperatures, high compressor stress and accumulated fatigue cycles scores higher, signalling that it is failing faster than the average engine at the same point in its operational history.

2.2 The PiNet Architecture

PiNet is a two-branch neural network that fuses temporal and instantaneous physics information through a shared embedding before producing its two outputs. The temporal branch operates on a thirty-cycle sliding window of the full eighty-nine engineered features, using a five-layer causal dilated convolutional stack [2] with dilation factors of 1, 2, 4, 8 and 16. These dilation factors give the branch a receptive field of sixty-three cycles, comfortably exceeding the window length and ensuring that the full recorded history available at any prediction point is accessible to the model. Output pooling concatenates the terminal and mean-pooled hidden states, yielding a one-hundred-and-twenty-eight-dimensional temporal representation⁴.

²See Appendix G, Eq. 1 for the full thermodynamic derivation and the regime-aware normalisation procedure applied to this feature.

³See Appendix G, Eq. 2 for the Cox proportional-hazards specification and the partial-likelihood estimation procedure.

⁴See Appendix A for the full parameter count by module, the receptive-field arithmetic, and the layer-normalisation conventions.

The temporal receptive field RF of the five-layer stack is given by the following standard formula for causal dilated convolutions (Equation 3), where $k = 3$ is the kernel size and d_i are the per-layer dilation factors:

$$RF = 1 + (k - 1) \times 2 \times \sum_i d_i = 1 + (3 - 1) \times 2 \times 31 = 63 \text{ cycles} \quad [\text{Eq. 3}]$$

The temporal branch can “see” up to 63 consecutive cycles of engine history when producing any single prediction — more than twice the 30-cycle input window. This extended reach lets the model detect slow-building degradation signatures that would be invisible in a shorter lookback window.

The physics branch is a compact two-layer network operating only on the last cycle of each window, taking as input the ten physics-specific features — including the compressor efficiency, the proportional-hazards score, the primary thermodynamic sensors, the cumulative-fatigue indices and the v0 health index — together with the operational regime weight that characterises the current flight condition. The physics branch produces a thirty-two-dimensional representation encoding the instantaneous physical state of the engine, independent of the temporal trajectory.

The fusion module combines the temporal and physics representations through a two-layer network producing a shared one-hundred-and-twenty-eight-dimensional embedding. These embedding drives two output heads: a regression head that produces a scalar Remaining Useful Life estimate and a classification head that assigns the observation to one of three operational risk bands — Healthy, Warning and Critical.

Table 3 — PiNet Structural Summary

Branch / Module	Inputs	Architecture	Output dimension
Temporal branch	30-cycle window × 89 features	5-layer causal dilated conv.	128
Physics branch	Last-cycle physics tokens + regime weight	2-layer network	32
Fusion module	Temporal (128) + Physics (32)	2-layer network	128 (shared embedding)
Regression head	Shared embedding (128)	2-layer regression	1 (RUL, cycles)
Classification head	Shared embedding (128)	2-layer classification	3 (risk-band logits)

Figure 1 provides a complete schematic of the PiNet architecture. The left branch processes the full 30-cycle sensor window through five dilated temporal convolutional layers; the right branch processes only the instantaneous physics tokens through a compact two-layer network. Both streams produce fixed-length vector representations which are combined by the fusion module before driving the two output heads.

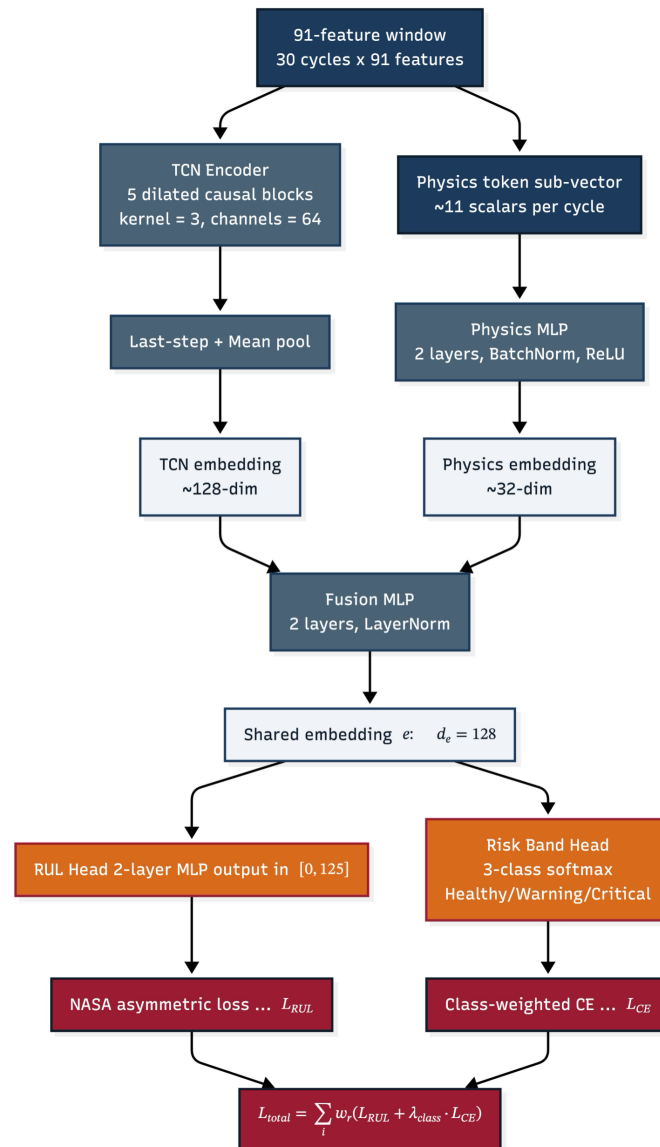


Figure 1 — PiNet architecture schematic. The temporal convolutional branch (left) captures degradation trajectory dynamics across 30 cycles. The physics branch (right) encodes instantaneous thermodynamic state. The fusion module combines both streams before the dual output heads produce a Remaining Useful Life estimate and a three-class operational risk assignment.

Figure 2 illustrates how the five dilated convolutional layers progressively extend the temporal receptive field. Each layer doubles the dilation factor, incorporating cycle history that is progressively further removed in time. At the output step (cycle $t = 29$) the highlighted path traces all inputs that contributed to the final representation.

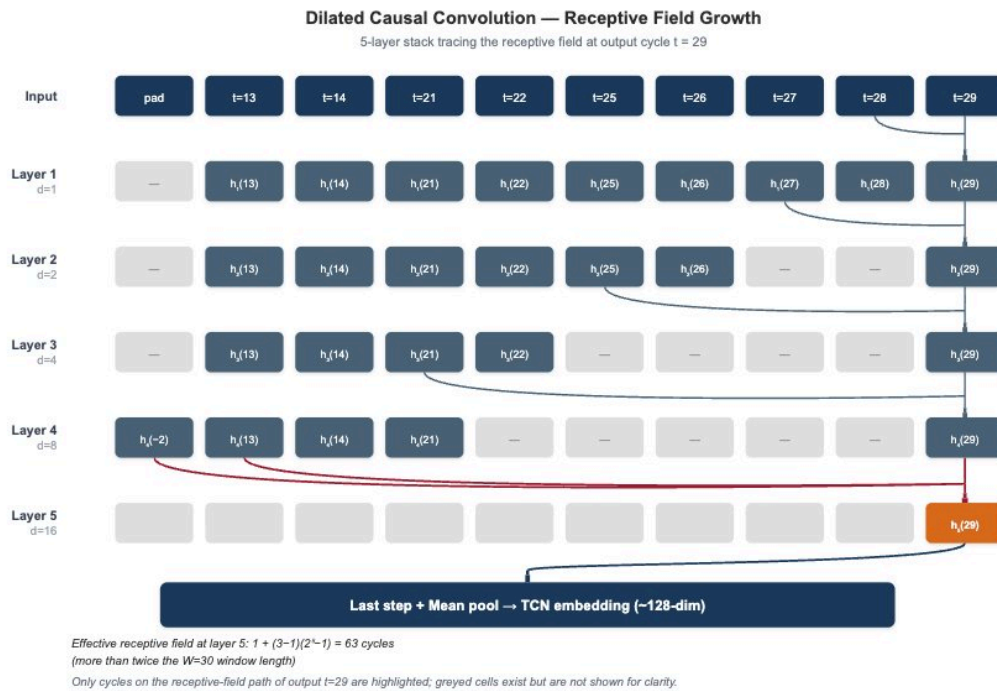


Figure 2 — Dilated causal convolution receptive field growth, traced at output cycle $t = 29$. The five-layer stack with dilation sequence $[1, 2, 4, 8, 16]$ achieves a receptive field of 63 cycles — more than twice the 30-cycle input window — allowing the model to detect both fast-onset and slow-building fault signatures.

3. Stage A Training and Convergence Behaviour

3.1 Training Configuration

Stage A training minimises a composite loss that combines a NASA asymmetric regression penalty with a cross-entropy companion on the risk-band head, both weighted by an inverse-frequency regime factor to ensure balanced representation across the six operational regimes present in the consolidated corpus⁵. The NASA asymmetric penalty is the scoring function published by Saxena and Goebel [1] that penalises late predictions more heavily than early predictions, reflecting the asymmetric operational cost of optimism in the face of degradation evidence relative to the cost of over-cautious early maintenance action⁶.

The full composite loss is defined by the following expression (Equation 4 — see Appendix B for the complete gradient-budget analysis):

$$L = E_i \left[w_r^{(i)} \left[s_i^{\text{NASA}} + \lambda_c CE_c(\hat{z}_i, y_i) \right] \right] \quad [\text{Eq. 4}]$$

The model is trained to minimise two kinds of mistakes simultaneously — getting the cycle-count prediction wrong and assigning the wrong urgency label. The regime weighting w_r ensures that flight conditions appearing less frequently in training still receive proportional attention. The balance

⁵See Appendix B, Eq. 4–Eq. 6 for the complete composite loss specification and the numerical dominance analysis.

⁶See Appendix C, Eq. 5–Eq. 6 for the NASA asymmetric scoring function and the penalty-ratio derivation across the operationally relevant cycle range.

between the two objectives proved problematic in Stage A (see Section 5.1 for the root-cause analysis).

Optimisation uses decoupled-weight-decay Adam [5] with a base learning rate of 1×10^{-3} and weight decay of 1×10^{-4} , scheduled over a planned eighty epochs with a single-epoch linear warmup followed by cosine annealing to zero. A stratified training sampler over-represents the scarce Critical band at a ratio of approximately 45:27.5:27.5 across Healthy, Warning and Critical to counteract the natural class imbalance. Gradient clipping is applied and early stopping halts training after twenty-five consecutive epochs without improvement on the internal-validation NASA score. The checkpoint with the lowest internal-validation NASA score is retained as the Stage A best model.

3.2 Convergence Outcome

Training converged to its best checkpoint at epoch three of the planned eighty, recording an internal-validation NASA score of 100,488, a Root Mean Square Error of 13.84 cycles and a Critical-band recall of 0.918. The run terminated by early stopping after eighteen epochs with no subsequent improvement. This extremely rapid convergence is not in itself unusual — deep networks trained with high learning rates can locate a low-loss minimum quickly — but the combination of a best epoch at three and monotonically worsening validation performance thereafter is the unambiguous signature of the optimiser having over-shot and then over-fitted for the remaining fifteen epochs.

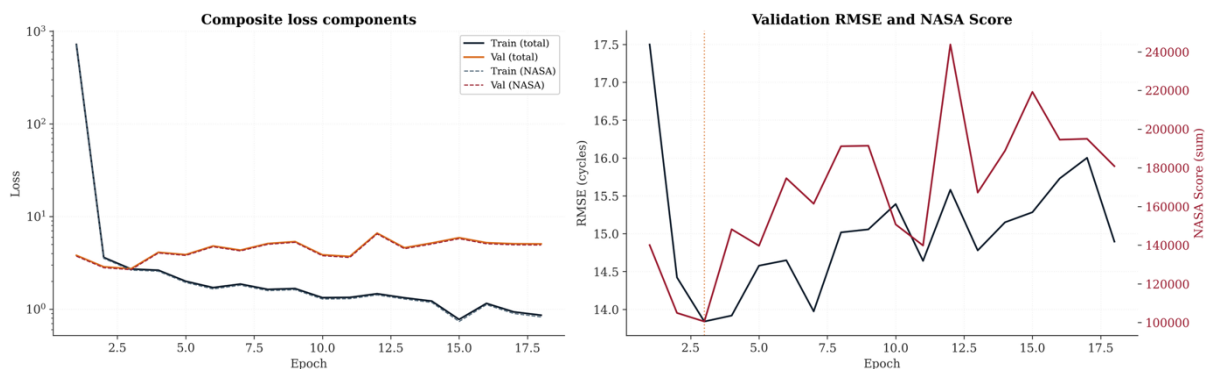


Figure 3 — Stage A training loss and validation score by epoch. Training loss (dark blue) descends rapidly; validation score (orange) reaches a minimum at epoch three and increases monotonically through epoch eighteen, the canonical over-fitting signature.

Figure 4 shows the per-class precision, recall and F_2 trajectories across the eighteen training epochs. The Critical-band recall (deep red) attains its peak at epoch three, coinciding with the best checkpoint, before deteriorating monotonically thereafter — the per-class signature of the same over-fitting pattern visible in the aggregate loss curves of Figure 1.

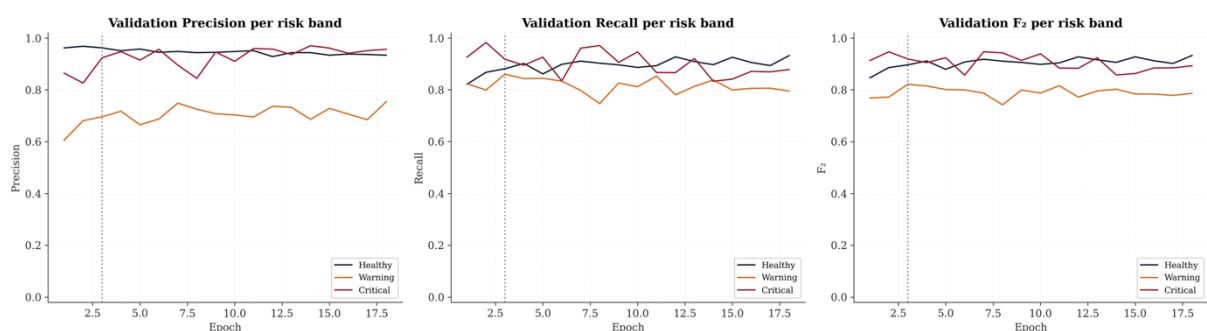


Figure 4 — Per-class precision, recall and F_2 on internal validation by epoch. Critical-band recall (deep red) peaks at epoch three (the best checkpoint) and deteriorates monotonically through epoch eighteen, the per-class over-fitting signature.

4. Iron Wall Headline Evaluation

4.1 Protocol

The NASA Official Test Set is opened exactly once under the Iron Wall Protocol. This protocol is the instrument that gives the comparative evidence its meaning: it ensures that test-set performance cannot be inflated by design choices made after observing test outcomes. Final-cycle predictions — one per engine — are used for all headline metrics.

4.2 Pooled Headline Result

The NASA Asymmetric Score is the official evaluation criterion for the C-MAPSS corpus, specified by Saxena and Goebel [1]. Its defining property is the deliberate asymmetry between the penalty for late predictions (over-optimism) and early predictions (over-caution), expressed by two separate formulas:

$$s_i^{\text{NASA}} = \exp\left(\frac{d_i}{10}\right) - 1, \quad d_i \geq 0 \quad [\text{Eq. 5}]$$

$$s_i^{\text{NASA}} = \exp\left(\frac{-d_i}{13}\right) - 1, \quad d_i < 0 \quad [\text{Eq. 6}]$$

If the model predicts too many cycles remaining ($d > 0$, over-optimistic), the penalty grows as $e^{\frac{d}{10}} - 1$ — accelerating rapidly. If it predicts too few ($d < 0$, over-cautious), the penalty is shallower: $e^{\frac{|d|}{13}} - 1$. A 30-cycle late prediction scores approximately 19 points; a 30-cycle early prediction scores approximately 10 points. This asymmetry deliberately encodes the operational reality that missing a genuine engine failure is more costly than scheduling maintenance early.

On every metric, PiNet Stage A performs materially worse than the v0 XGBoost reference (Table 4). The pooled Root Mean Square Error increases by 9.81 cycles and the NASA Asymmetric Score — the primary aggregate measure of operational risk, which penalises late predictions exponentially more than early predictions — is 7.4 times larger. The Critical-band recall falls from 0.9363 to 0.4204, a reduction of 0.5159 units.

How to read the table. Lower RMSE and NASA scores are always better. Higher recall and F_2 are always better. A model that achieves a low RMSE but misses failing engines is not safer — it is less safe. Critical-Band Recall and the Safety Gate are therefore the binding metrics in this programme; RMSE and MAE provide supporting context on raw prediction accuracy.

Table 4 — Pooled Headline Metrics: v0 XGBoost versus PiNet Stage A

Metric	v0 XGBoost	PiNet Stage A	Δ
Root Mean Square Error (cycles)	14.85	24.66	+9.81
Mean Absolute Error (cycles)	10.41	18.83	+8.42
NASA Asymmetric Score	4,336	32,204	+27,868 (7.4×)
Critical-Band F_2 Score	0.9339	0.4735	−0.4604
Critical-Band Recall	0.9363	0.4204	−0.5159

4.3 Subset Breakdown

Table 5 decomposes the result by C-MAPSS subset. The performance gradient across subsets is strongly structured by operational complexity. On the single-fault, single-regime subset the Critical-band recall reaches 0.640 — below the v0 reference but workable. On the dual-fault, six-regime subset it

collapses to 0.115, and this one subset alone contributes sixty-seven per cent of the pooled NASA score, overwhelming the more modest failures on the simpler subsets. The dual-fault, six-regime combination is therefore the test case that most discriminates between v0 and v1.

Table 5 — Per-Subset Stage A Performance

Subset	Fault Mode	Regime Complexity	n	RMSE	NASA	F_{2c}	R_c
FD001	Single fault (HPC)	Single regime	100	16.13	502	0.684	0.640
FD002	Single fault (HPC)	Six regimes	259	20.95	4,930	0.588	0.533
FD003	Dual fault	Single regime	100	25.76	5,160	0.645	0.600
FD004	Dual fault	Six regimes	248	30.04	21,612	0.140	0.115

4.4 The Critical-Recall Safety Gate

The Critical-Recall Safety Gate is the binding pass-or-fail criterion for deployment consideration. It requires that the PiNet Critical-band recall not fall more than two percentage points below the v0 reference. The gate condition and the measured result are stated below.

$$\Delta R_c = R_c^{\text{PiNet}} - R_c^{v0} \geq -0.02 \quad [\text{Eq. 7}]$$

$$\Delta R_c = 0.4204 - 0.9363 = -0.5159 \quad [\text{Eq. 8}]$$

Critical-Recall Safety Gate: X FAIL

4.5 What the Evaluation Diagnostics Show

The diagnostic visualisations from the headline evaluation tell a consistent story. The predicted-versus-actual scatter plot (Figure 5) exhibits a pronounced "banana" shape: PiNet predictions cluster substantially above the equality line at low true Remaining Useful Life, the direct geometric signature of systematic late-prediction bias on engines that are genuinely near failure. The per-band residual breakdown (Figure 6) confirms that Critical and Warning bands carry overwhelmingly positive residual mass — the model over-predicts Remaining Useful Life precisely where it is most dangerous to do so. The per-engine trajectory overlays (Figure 7) extend this picture across all four subsets, showing predicted degradation curves that lag the true trajectories throughout the P-F interval.

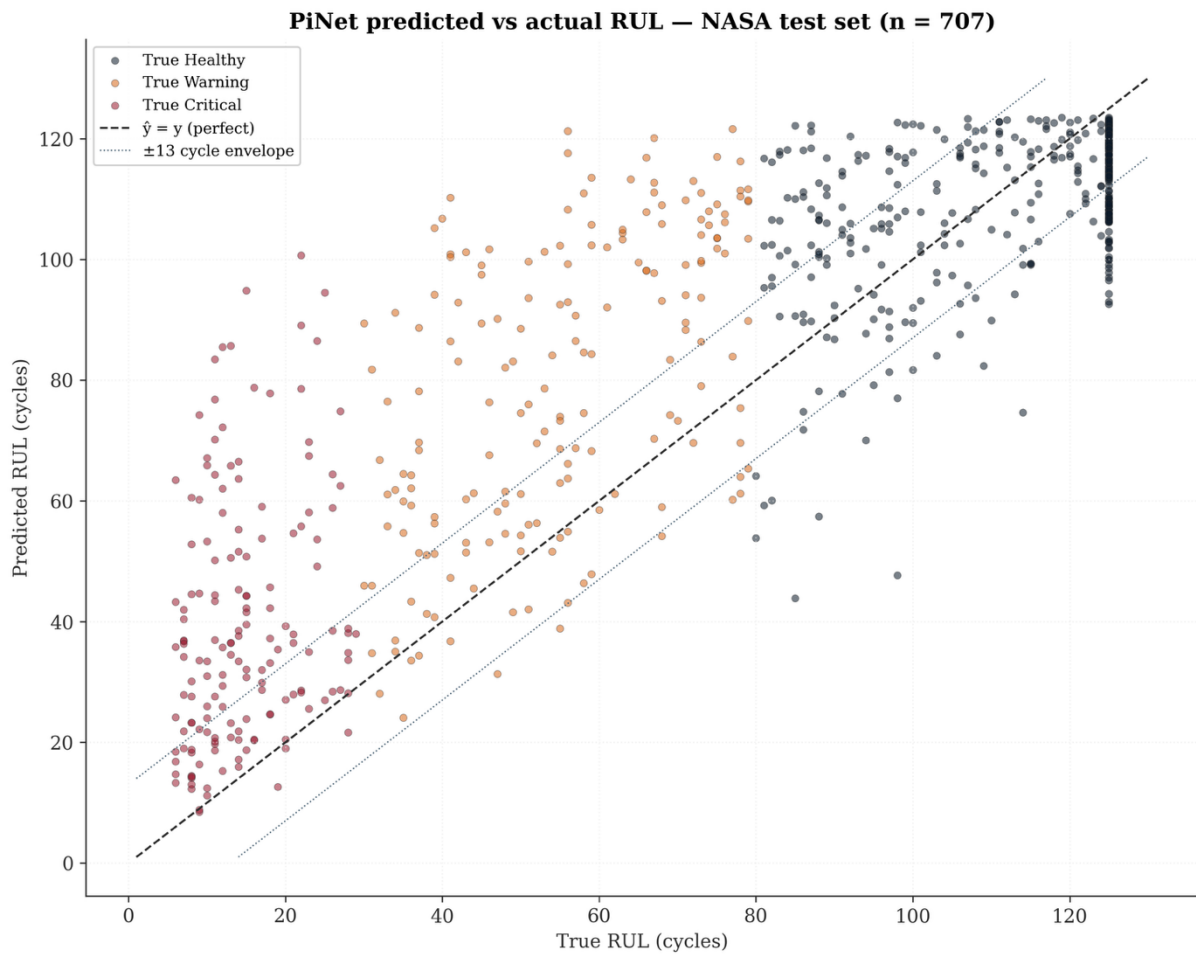


Figure 5 — Predicted versus actual Remaining Useful Life, coloured by true risk band. The systematic over-prediction at low true RUL is the operational cost of the safety-gate failure.

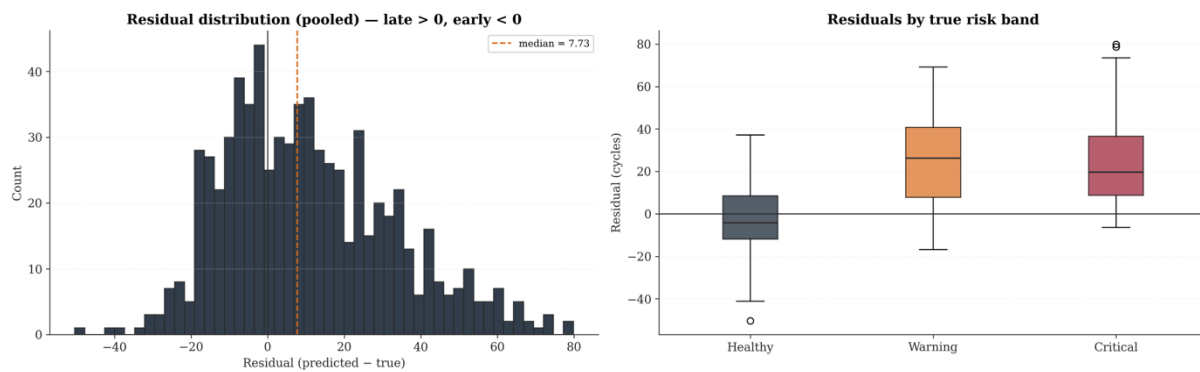


Figure 6 — Left: pooled residual histogram with median annotation. Right: per-band residual box-plot. The Critical (deep red) and Warning (orange) bands show overwhelmingly positive residual mass — the model over-predicts Remaining Useful Life exactly where it is most dangerous to do so.

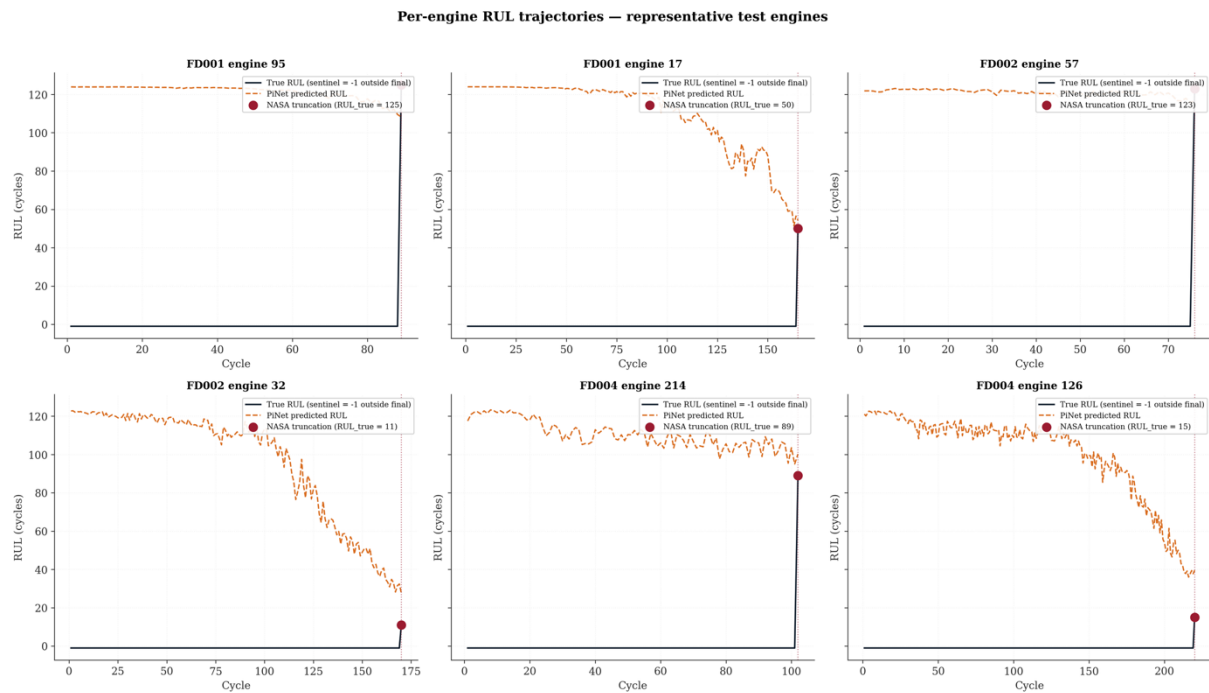


Figure 9 — Predicted degradation trajectory overlays for representative engines across all four C-MAPSS subsets. Predicted curves visibly lag the true trajectories throughout the P-F interval, with the lag most pronounced on dual-fault engines — the same structural weakness that drives the headline safety-gate failure.

The three-class confusion matrix (Figure 8) translates the recall number into an operational decision context. The Critical-band row, row-normalised to show recall rates, confirms that approximately fifty-eight per cent of engines that are genuinely in the Critical band are assigned to the Warning band. In a maintenance scheduling context this means that more than half of the fleet's most urgent cases would receive a lower escalation priority than their condition warrants.

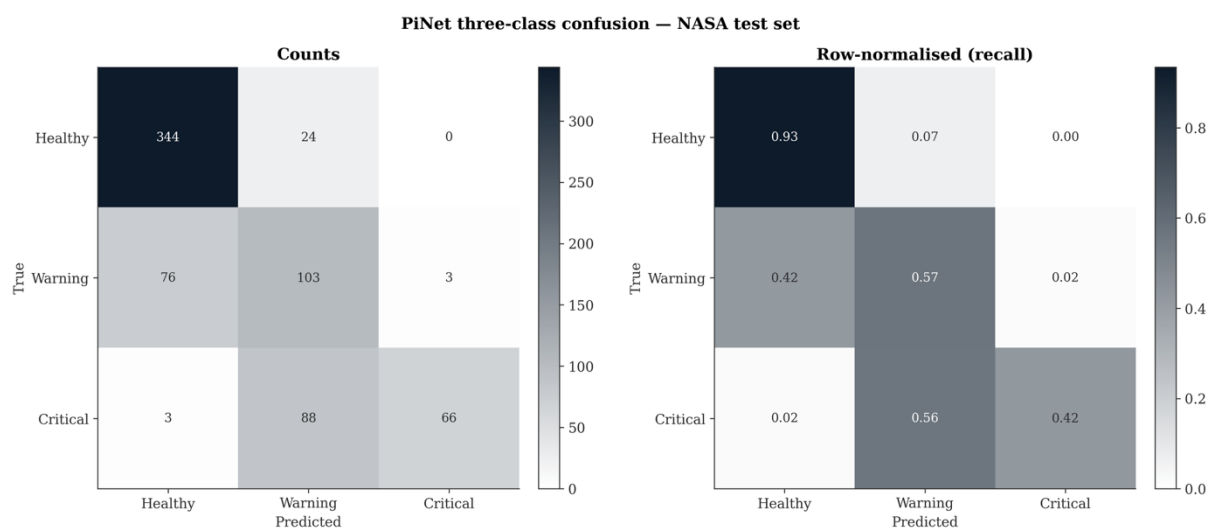


Figure 11 — Three-class confusion matrix (row-normalised, showing recall). Critical-band row confirms 58% of true-Critical engines are mislabelled as Warning.

5. Operational Interpretation

5.1 The Four Root Causes

The Stage A failure is not attributable to an architectural flaw in PiNet. The architecture itself is sound and its internal validation metrics during training — particularly the Critical-band recall of 0.918 at epoch three — demonstrate that the model is capable of learning the relevant discriminative patterns. The failure is attributable to four compounding training-configuration deficiencies, each independently correctable and each arising from choices that interact adversely.

The first cause is an excessive learning rate. At 1×10^{-3} , the optimiser reaches a low-loss region of the parameter space in three training epochs and then over-fits progressively through the remaining fifteen before early stopping intervenes. A more conservative learning rate would extend the productive training horizon and improve the probability of settling in a flatter, better-generalising minimum.

The second is insufficient regularisation on the temporal branch. The temporal branch accounts for approximately eighty-three per cent of the total parameter count. Against an aggressive learning rate, the regularisation applied is not sufficient to prevent the temporal branch from memorising training-set trajectories rather than generalising degradation behaviour.

The third is a distributional mismatch introduced by the stratified training sampler. By ensuring that every training minibatch contains a disproportionately high fraction of Critical-band observations, the sampler creates a training distribution that does not match the natural evaluation distribution, in which Critical engines constitute only about sixteen per cent of the cohort. The model learns a prediction policy optimised for the inflated-Critical training distribution, producing systematic late-prediction bias when evaluated on the natural test distribution. The quantitative consequence is a near-doubling of mean absolute cycle-level error (18.83 versus 10.41 cycles for v0), driven almost entirely by this mismatch⁷.

The fourth is numerical dominance of the regression objective over the classification objective. The NASA asymmetric penalty grows exponentially with the cycle-level prediction error, whilst the cross-entropy classification loss is bounded in a narrow range. At the operating conditions where the headline metrics are most sensitive, the regression gradient is approximately thirteen times larger than the classification gradient, giving the classification head insufficient gradient budget to learn effectively⁸.

5.2 Implications for Fleet Management

The failure mode diagnosed above is, in a fleet management context, the most operationally significant category of predictive maintenance error. A system that under-detects imminently failing engines does not merely fail to add value: it actively creates risk by providing false reassurance that can delay maintenance escalation on engines that require it. The v0 XGBoost baseline, with its Critical-band recall above 0.93, is consistent with the dispatch-reliability and aircraft-on-ground cost-avoidance objectives of the FusionCore commercial case. PiNet Stage A, in its present configuration, is not. The Stage B remediation pathway is well-defined and addresses each of the four causes with targeted, proportionate adjustments.

6. Feature Retention Decision — Sensor 16

A deferred decision from the v0 programme required v1 to determine whether sensor s_{16} (the fan-bypass stream signal) carries sufficient information to justify its retention in the feature manifold. A statistical audit combining within-regime variance and rank-correlation analysis was conducted on the internal validation cohort. The audit found that sensor s_{16} is effectively inactive in five of the six operational

⁷See Appendix E for the full quantitative analysis of the distribution mismatch and the recommended Stage B remedy.

⁸See Appendix D for the gradient-budget analysis, including the computed dominance ratio at the operationally critical cycle-error range.

regimes but carries statistically significant information in the high-altitude regime, where its rolling-mean variant correlates with the prediction residual at a rank-correlation coefficient of 0.297 with a p-value of approximately 7×10^{-16} . The two-panel statistical audit is shown in Figure 7. The retention decision is therefore **retain**⁹.

The practical implication is that any engineering team considering removing sensor s_{16} from the instrumentation or preprocessing pipeline for operational simplicity should be aware that doing so would degrade prediction accuracy specifically on high-altitude operating points, the regime in which the signal is most active.

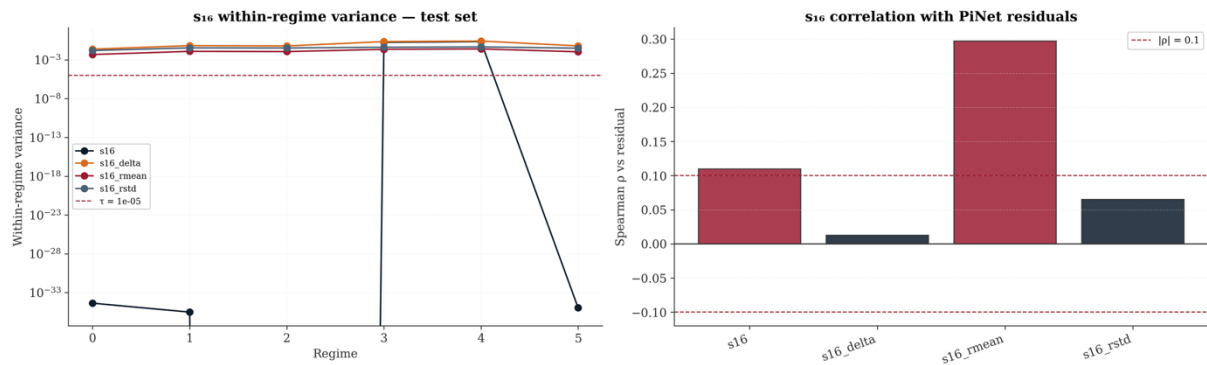


Figure 9 — Two-panel sensor s_{16} statistical audit. Left: per-regime variance for the raw, delta, rolling-mean and rolling-standard-deviation variants. Right: Spearman rank correlation between each variant and the PiNet prediction residual on final-cycle observations. The rolling-mean variant carries the dominant signal in the high-altitude regime.

7. Pre-Declared Metric-Learning Ablation

7.1 Objective and Design

A metric-learning ablation was pre-declared in the v1 programme specification as a conditional experiment to assess whether augmenting the composite training loss with an embedding-geometry regulariser would improve the separability of the three risk bands in the shared representation. Two additional loss terms were introduced: a triplet margin loss on the shared embedding [6] that directly incentivises distance-based separation between risk classes, and a Shannon marginal-entropy regulariser that discourages representation collapse. The augmented loss was applied to a stratified thirty per cent subset of the training cohort over twenty additional training epochs from the Stage A best checkpoint.

⁹See Appendix F, Table F.1 for the full variance and correlation diagnostics by regime.

The mechanism of the triplet loss is illustrated in Figure 10. Three observations are processed simultaneously through the same shared encoder: an anchor engine engine window, a positive example from the same risk band, and a negative example from a different band. The loss function penalises the model whenever the anchor-to-positive margin distance in embedding space exceeds the anchor-to-negative distance by less than a fixed safety margin $\alpha = 0.5$.

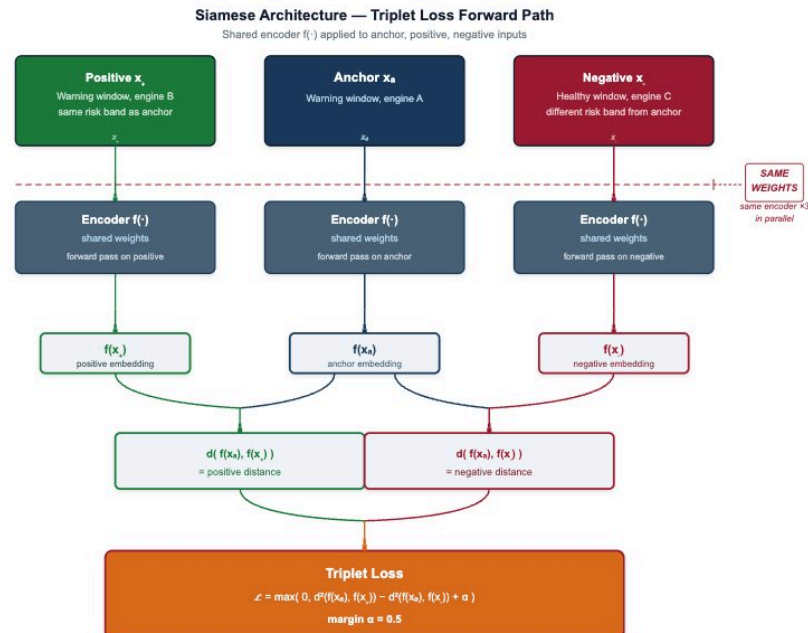


Figure 10 — Siamese architecture for the triplet loss ablation. The encoder $f(\cdot)$ is applied in parallel to anchor, positive and negative observations using shared weights. The loss directly optimises embedding geometry so that same-band observations sit closer together than different-band observations, with margin $\alpha = 0.5$.

The network is rewarded for organising its internal “map” of engines so that engines in the same urgency category cluster together, and engines in different categories are pushed apart. This encourages the shared representation to encode risk-band identity as well as raw cycle-count prediction, improving the separability of the three operational risk classes in the model’s learned space.

7.2 Ablation Result

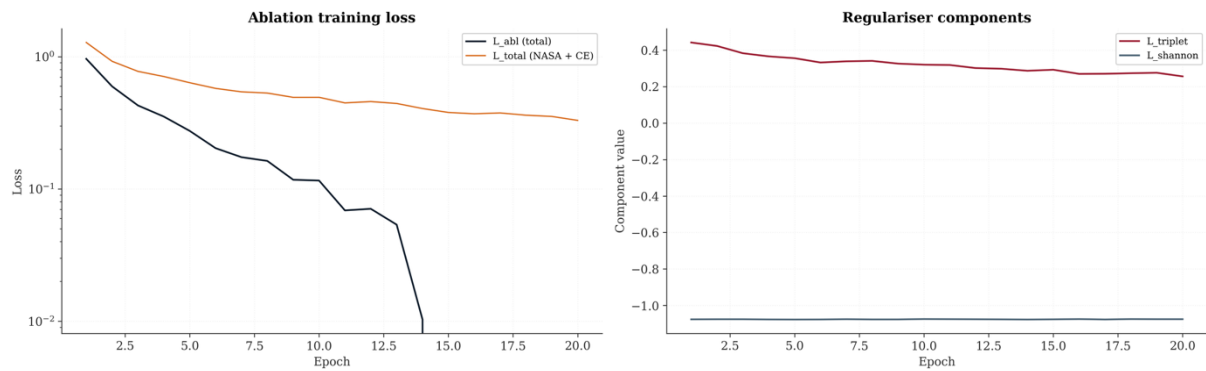


Figure 11 — Augmented composite loss components by ablation epoch: total loss (dark blue), base NASA plus cross-entropy loss, triplet margin loss and Shannon entropy regularisation term. All four components descend cleanly, confirming that the augmented loss is numerically well-conditioned throughout the twenty ablation epochs.

The ablation training dynamics are shown in Figure 11, confirming that all four loss components descend cleanly across the twenty epochs. The embedding separation, measured by the Kolmogorov–Smirnov statistic on the distance distributions of same-class versus different-class embedding pairs [7], improved from 0.661 to 0.937, an increase of 0.276 with p-values approaching zero in both cases. By the pre-declared decision criterion this constitutes a pass, and the visual separation of the distance distributions confirms that the metric-learning objective has succeeded in the geometric sense (Figure 12).

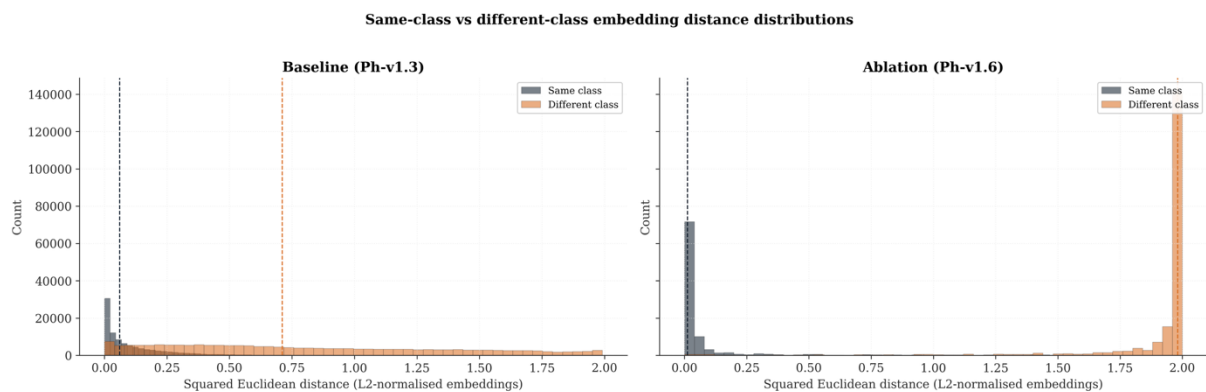


Figure 12 — Same-class (dark blue) versus different-class (deep red) pairwise embedding distances: baseline (left) and ablation (right). The ablation exhibits a substantially wider bimodal separation, confirming the geometric improvement.

Table 6 — Ablation Embedding-Separation Result

Model	KS statistic	p-value	Decision
Stage A baseline	0.661	≈ 0	Reference
Ablation (triplet + entropy losses)	0.937	≈ 0	Improved
Delta	+0.276	—	—

7.3 Why the Rollout Is Qualified

A strong Kolmogorov–Smirnov separation metric is a necessary but not sufficient condition for adoption. Embedding-geometric improvement does not imply that the Critical-Recall Safety Gate will pass: a well-separated embedding can still drive a regression head that systematically over-predicts Remaining Useful Life. Two additional fragilities were identified during the ablation: the entropy regulariser showed parameter sensitivity under the stratified sampling distribution, and the KS-only adoption criterion is insufficiently rich for a safety-critical deployment decision. Adoption of the metric-learning augmentation is therefore recommended only in combination with the Stage B parametric adjustments of Section 8.

8. Recommendations and Forward Path

8.1 Stage B Retraining Specification

Table 7 summarises the parametric adjustments recommended for Stage B. Each adjustment maps directly to one of the four root causes of the Stage A failure and is calibrated to be proportionate. Taken together they address the learning-rate, regularisation, sampler and gradient-budget dimensions simultaneously, which is necessary because the four causes interact: correcting any one in isolation leaves the others in place.

Table 7 — Stage B Recommended Parametric Adjustments

Root Cause	Stage A Configuration	Stage B Specification
Excessive learning rate	1×10^{-3}	3×10^{-4}
Insufficient regularisation	Dropout 0.15 (temporal branch)	Dropout 0.25 (temporal branch)
Sampler distribution mismatch	Stratified 45:27.5:27.5 sampling	Natural distribution; class-weighted cross-entropy retained
Regression-gradient dominance	NASA per-row loss unscaled	NASA per-row loss scaled by factor 1/50 to 1/100

8.2 Probabilistic Calibration

A separate limitation of the present v1 output is the absence of uncertainty quantification on the Remaining Useful Life estimate. Point estimates without confidence intervals constrain operational use in maintenance planning, dispatch scheduling and risk-based inspection. A post-training probabilistic calibration phase is therefore recommended for any Stage B continuation: split-conformal prediction intervals on the regression head and temperature scaling on the risk-band classifier are well-suited to the architecture, are post-hoc in nature and do not require retraining the backbone.

8.3 Decision

It is recommended that PiNet Stage A be retained as recorded evidence base and that no Stage A output be considered for operational deployment. The v0 XGBoost baseline remains the reference model for all operational purposes. Stage B retraining against the adjustments in Table 7 is the proposed immediate forward action. The pre-declared metric-learning ablation result is recommended for adoption exclusively in combination with the Stage B specification.

Bibliography

- [1] Saxena, A. and Goebel, K. (2008). Turbofan engine degradation simulation data set. NASA Ames Prognostics Data Repository.
- [2] Bai, S., Kolter, J. Z. and Koltun, V. (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv:1803.01271.
- [3] Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society B*, 34(2), 187–220.
- [4] Heimes, F. O. (2008). Recurrent neural networks for remaining useful life estimation. *Proceedings of the International Conference on Prognostics and Health Management, PHM 2008*.
- [5] Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. *International Conference on Learning Representations, ICLR 2019*.
- [6] Schroff, F., Kalenichenko, D. and Philbin, J. (2015). FaceNet: a unified embedding for face recognition and clustering. *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2015*.
- [7] Massey, F. J. (1951). The Kolmogorov–Smirnov test for goodness of fit. *Journal of the American Statistical Association*, 46(253), 68–78.
- [8] Walsh, P. P. and Fletcher, P. (2004). *Gas Turbine Performance*, 2nd edition. Blackwell / Wiley.
- [9] Saxena, A. et al. (2008). Damage propagation modeling for aircraft engine run-to-failure simulation. *Proceedings of the International Conference on Prognostics and Health Management, PHM 2008*.

Appendix A — PiNet Architecture Specification

The parameter count by module is presented in Table A.1. The temporal branch accounts for the overwhelming majority of trainable parameters, which is consistent with its role in learning long-horizon degradation patterns across the full eighty-nine-feature input space over a thirty-cycle window. The physics branch is deliberately compact, reflecting the intent that it should encode instantaneous state rather than time series.

Table A.1 — PiNet Parameter Count by Module

Module	Parameters	Notes
Temporal branch block 1 (dilation 1)	~30,500	Includes residual projection layer
Temporal branch block 2 (dilation 2)	~24,800	—
Temporal branch block 3 (dilation 4)	~24,800	—
Temporal branch block 4 (dilation 8)	~24,800	—
Temporal branch block 5 (dilation 16)	~24,800	—
Physics branch	~2,200	10 physics tokens + regime weight input
Fusion module (160 → 144 → 128)	~44,500	—
Regression head (128 → 64 → 1)	~8,300	Scalar RUL output
Classification head (128 → 64 → 3)	~8,400	Three risk-band logits
Total	~193,000	Final trained parameter count

The temporal receptive field of the five-block stack spans sixty-three cycles, derived from the standard causal dilated convolution arithmetic with kernel size three and dilation sequence 1, 2, 4, 8, 16. This exceeds the thirty-cycle window, providing complete coverage of all available history at every prediction point.

Appendix B — Composite Loss and Gradient Dynamics

The composite loss minimised during Stage A training is the regime-weighted expectation over minibatch rows of the sum of the NASA asymmetric regression term and a scaled cross-entropy classification companion. The full expression is given below.

$$L = E_i \left[w_r^{(i)} \left[s_i^{\text{NASA}} + \lambda_c \text{CE}_c(\hat{z}_i, y_i) \right] \right] \quad [\text{Eq. 4}]$$

In this expression, the superscript (i) denotes the row index; the regime weight w_r is the inverse frequency of the regime in the training corpus, normalised to sum to unity; λ_c is the cross-entropy scaling factor, fixed at 0.3 throughout Stage A; and the cross-entropy term CE is evaluated on the risk-band classification head output against the true risk-band label. The NASA asymmetric penalty s_i is defined by Equations 5 and 6 below.

$$s_i^{\text{NASA}} = \exp\left(\frac{d_i}{10}\right) - 1, \quad d_i \geq 0 \quad [\text{Eq. 5}]$$

$$s_i^{\text{NASA}} = \exp\left(\frac{-d_i}{13}\right) - 1, \quad d_i < 0 \quad [\text{Eq. 6}]$$

The variable d_i denotes the signed prediction error (predicted minus true Remaining Useful Life in cycles). The asymmetric constants 10 and 13 are the values published by Saxena and Goebel [1] and are not free hyperparameters. The lower constant in the late-prediction branch ($d \geq 0$) produces more rapid growth and thereby encodes the operational principle that optimism about remaining life is more dangerous than pessimism.

B.1 Gradient Budget Dominance

The practical consequence of the unbounded exponential growth in the NASA term is that the regression gradient overwhelms the classification gradient in any minibatch containing rows with non-trivial cycle-level prediction errors. At a cycle error of thirty — the typical order of magnitude for an untrained or poorly calibrated model on the Critical band — the per-row NASA penalty is approximately $e^3 - 1$, which equals approximately 19. The corresponding cross-entropy contribution for a confidently misclassified Critical observation, multiplied by the scaling factor $\lambda_c = 0.3$, is approximately 1.5. The resulting regression-to-classification gradient ratio is approximately 13:1. This dominance explains why the Stage A Critical-band recall is low despite the fact that the classification head architecture is itself capable of producing high recall (as evidenced by the 0.918 achieved at epoch three). The Stage B recommendation to scale the NASA term by a factor between 1/50 and 1/100 is calibrated to bring the two contributions to comparable numerical magnitude in the operating region of interest.

Appendix C — NASA Asymmetric Scoring Function

The NASA asymmetric scoring function, introduced by Saxena and Goebel [1] as the standard evaluation criterion for the C-MAPSS corpus, is defined by Equations 5 and 6 of Appendix B. The total score over the test set is the sum of per-engine final-cycle values. A lower score is better; the minimum achievable score for a perfect predictor is zero.

The penalty-ratio between a late prediction and an equal-magnitude early prediction grows from approximately 1.3× at small errors to approximately 9.2× at a cycle error of one hundred. This asymmetry reflects the engineering judgement that aircraft-on-ground and airworthiness risk arising from an undetected impending failure is substantially greater than the cost of scheduling maintenance too early. For a commercial aviation operator, the mapping is direct: a late prediction on a Critical engine risks an in-service rejection or unscheduled removal, both of which are substantially more expensive per event than a conservative early removal.

Appendix D — Gradient Budget Analysis

Table D.1 quantifies the NASA-to-cross-entropy gradient ratio as a function of the per-row cycle error, evaluated at $\lambda_c = 0.3$ and assuming a maximum-confidence misclassification (cross-entropy $\approx 5 \times 0.3 = 1.5$). The dominance becomes operationally significant above a cycle error of approximately twenty, which is well within the typical prediction-error range at the start of Stage A training.

Table D.1 — NASA-to-Cross-Entropy Gradient Ratio by Cycle Error

Cycle error d	NASA per-row penalty	CE per-row ($\lambda_c = 0.3$)	Gradient ratio
5	0.65	1.50	0.4×
10	1.72	1.50	1.1×
20	6.39	1.50	4.3×
30	19.1	1.50	12.7×
50	147	1.50	98×
100	22,026	1.50	14,684×

Appendix E — Training Sampler Analysis

The stratified training sampler constructs each minibatch with the Healthy, Warning and Critical bands at a ratio of 45:27.5:27.5. This ensures that every minibatch contains approximately twenty-nine Critical-band observations, irrespective of their natural frequency in the training corpus. The objective is to provide the model with sufficient Critical-band gradient signal to learn the decision boundary in the presence of class imbalance.

The unintended consequence is that the model learns a prediction policy calibrated to a Critical-band frequency of 27.5 per cent. In the NASA Official Test Set, the true Critical-band frequency is approximately 16 per cent. The model therefore over-predicts Critical membership under the training distribution, which translates into systematic over-prediction of the number of cycles remaining when evaluated on the test distribution. This is the distributional mismatch mechanism.

The quantitative signature is the near-doubling of mean absolute cycle error: 18.83 cycles for Stage A versus 10.41 cycles for the v0 baseline. The Stage B remedy is to remove the stratified sampler and train on the natural distribution. The class-imbalance problem it was intended to address is handled instead by the class-weighted cross-entropy term, which provides the same gradient-emphasis effect without distorting the marginal distribution of predictions.

Appendix F — Sensor s_{16} Statistical Audit

The audit applies two retention criteria to three derived variants of sensor s_{16} (the fan-bypass stream signal): the raw value, a rolling mean and a rolling standard deviation. A sensor variant is retained if its maximum within-regime variance exceeds 1×10^{-5} or if its absolute rank-correlation with the PiNet prediction residual exceeds 0.10. Table F.1 reports the results.

Table F.1 — Sensor s_{16} Retention Audit Results

Diagnostic	Threshold	Measured value	Decision
Maximum within-regime variance (any variant)	$> 1 \times 10^{-5}$	0.266 (regime 4, rolling mean)	Retain
Maximum absolute Spearman correlation with prediction residual	> 0.10	0.297 (rolling mean)	Retain
Significance (p-value at maximum correlation)	< 0.05	$\approx 7 \times 10^{-16}$	Retain

The rolling-mean variant is the load-bearing form: it smooths the bimodal excursions between high-altitude and low-altitude phases of the flight cycle that collapse the information content of the raw column. The audit outcome is unambiguous across all three criteria.

Appendix G — Compressor Efficiency and Proportional-Hazards Derivations

G.1 Compressor Isentropic Efficiency

The compressor isentropic efficiency is derived from the ideal-gas isentropic relations applied across the high-pressure compressor stage, following the standard treatment in Walsh and Fletcher [8]. It is the ratio of the ideal isentropic enthalpy rise — computed from the measured inlet conditions and the pressure ratio — to the actual enthalpy rise measured across the stage.

$$\eta_c = \frac{T_2 \left[\left(\frac{P_{30}}{P_2} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]}{T_{30} - T_2} \quad [\text{Eq. 1}]$$

In this expression, T_2 and T_{30} are the compressor inlet and outlet total temperatures; P_2 and P_{30} are the corresponding total pressures; and $\gamma = 1.4$ is the ratio of specific heats for air under the relevant operating conditions. The expression is evaluated in physical units (Kelvin, consistent pressure units) before any statistical normalisation is applied, ensuring that the derived quantity has a thermodynamically meaningful interpretation as a health proxy. It is subsequently normalised within each operational regime using parameters estimated exclusively from the training partition.

G.2 Cox Proportional-Hazards Score

The proportional-hazards model, introduced by Cox [3], specifies the conditional hazard of engine failure as the product of a non-parametric baseline hazard and a parametric exponential term that depends on a vector of covariates. The canonical form is as follows.

$$\lambda(t | x) = \lambda_0(t) \exp(\beta^T x) \quad [\text{Eq. 2}]$$

Here $\lambda_0(t)$ is the unspecified baseline hazard at cycle t , β is the coefficient vector estimated by Cox partial likelihood [3], and x is the covariate vector for a given engine-cycle observation. The partial-hazard ratio $\exp(\beta^T x)$ is what enters the architecture as a physics feature: it quantifies the instantaneous failure-risk multiplier for that observation relative to the baseline. The partial-likelihood approach circumvents the need to specify λ_0 explicitly, making the method tractable for the present application. The coefficient vector β is estimated on the internal training cohort using ten covariates: the primary thermodynamic sensors s_4 (exhaust gas temperature), s_7 and s_9 ; the compressor pressure ratio; the cumulative thermal fatigue index; three cumulative-fatigue derived features; the corrected core speed ratio; and the v0 composite health index.